

Report: Quantitative GWAS of barley breeding

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1 Summary

About

- I run GAPIT on the quantitative barley traits and the genotype data from a barley population
- I also implemented the core functionality of a GWAS, especially for understanding the linear regression of the markers

Data

- This data was provided by the MLU Halle during my master programme
- Quantitative phenotype traits and a genotype matrix for multiple breed lines and multiple markers of Chromosome 2 in barely

HEA	= Heading date	(Ährenschieben)
TGW	= Thousand Grain Weights	(Tausendkorngewicht)
HEI	= Height	(Höhe)
RIP	= Ripening time	(Reifedauer)
PM	= Powdery Mildew Susceptibility	(Mehltauafälligkeit)

Quantitative Traits (Phenotype)

- Normally distribution of HEA, TGW, HEI and RIP (potential poly genetic orchestration)
- Bimodal distribution of PM, hint for a genetically distinct groups in the population
- Strong anti correlation between HEA and RIP
- Weak correlation between TGW and RIP
- Weak correlation between TGW and HEA

2 Phenotype

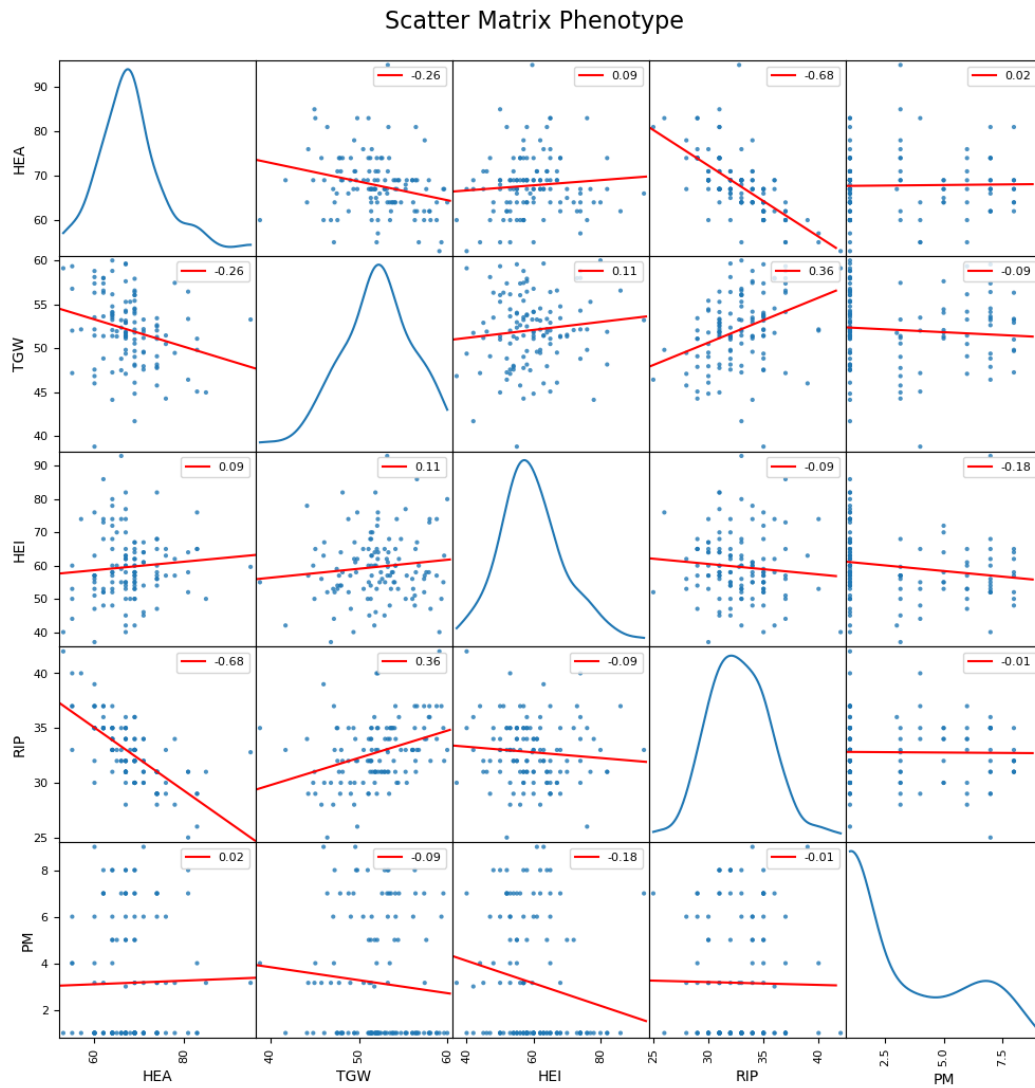


Fig. 1: Scatter matrix of phenotype data.

KDE

- All properties beside the PM are almost normally distributed (uni-modal)
- These properties seems to undergo a complex genetic orchestra
- Instead, PM shows a bimodal shaped distribution, hence one might argue that there exist two distinct populations (or genetic patterns), the majority has a strong resistance where a minority is more susceptible

TGW $\times$ RIP

- If the desired trait is a greater crop (yields more food), one can clearly see that TGW (Thousand Grain Weight) slightly correlates with the RIP (Ripening Time)
- The crop might have more time accumulating consumable nutrients (sugars, proteins, lipids)
- The scatter also shows that one would select crops that ripen later

TGW $\times$ HEA

- TGW slightly anti-correlates with HEA (Heading Date)
- But the scatter doesn't show that very clearly
- One might select for early headers

HEA $\times$ RIP

- Strong anti correlation
- High chance of a co-regulated gene orchestration

HEI $\times$ PM

- Maybe not relevant for breeding, but interesting behavior of barley, when they have weaker resistance against powder mildew, the crops heights reduce variance and shrink

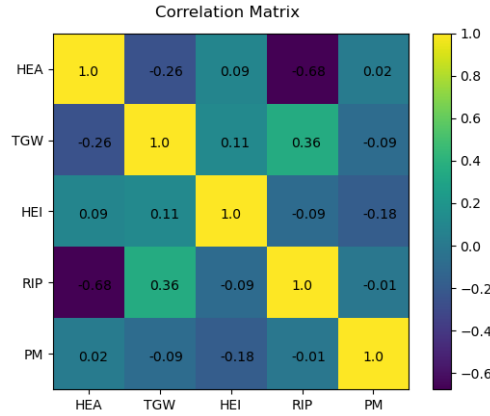


Fig. 2: Correlation matrix of phenotype data.

3 GWAS

3.1 Model Evaluation

Since I used two different models, I compare the QQ plots to pick the best model for this data.

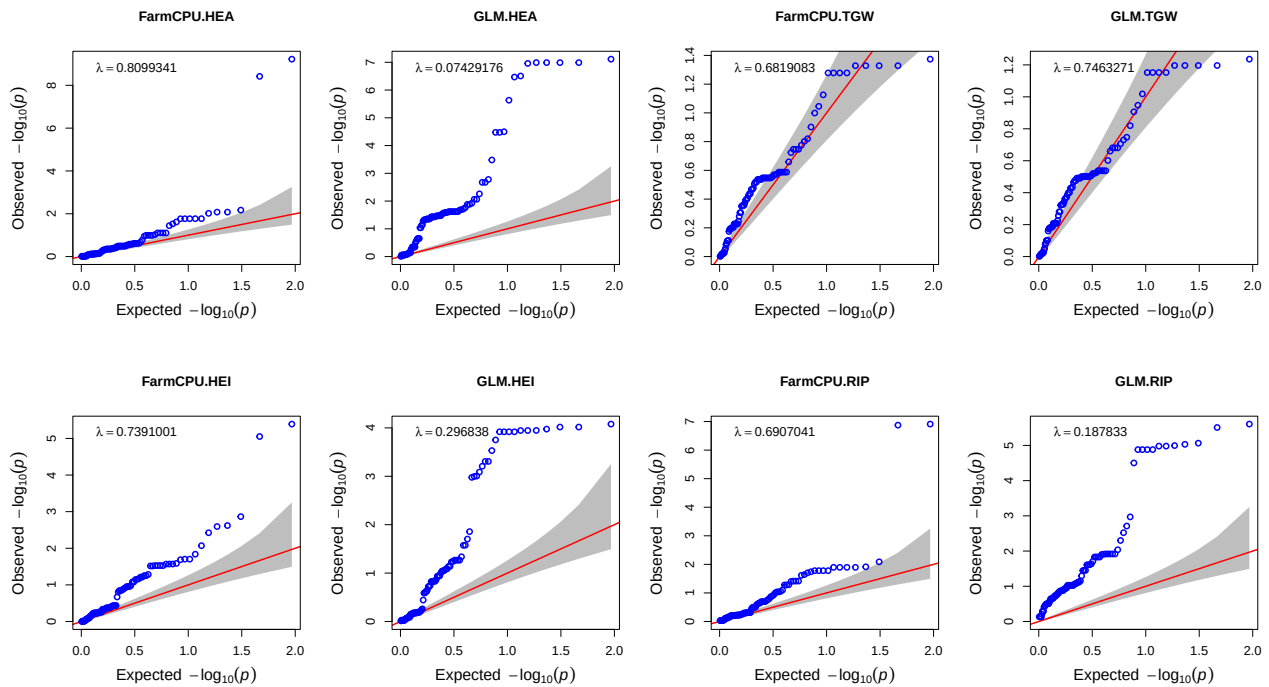


Fig. 3: Model Evaluation (FarmCPU & GLM) of HEA, TGW, HEI and RIP.

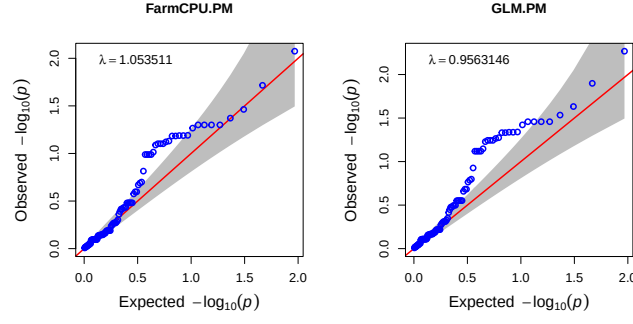


Fig. 4: Model Evaluation (FarmCPU & GLM) of PM

- FarmCPU outperforms the GLM w.r.t. HEA, HEI, RIP
- Both models perform equally w.r.t. TGW and PM
- The GLM shows multiple observed p values that doesn't align with the expected (false-positives), but FarmCPU shows just a few outlier, this indicates that they are supposedly true-positives
- Therefore, I use the FarmCPU as reference

3.2 Manhattan Plots

- Visualizes genome-wide associations of traits of interest, given the model

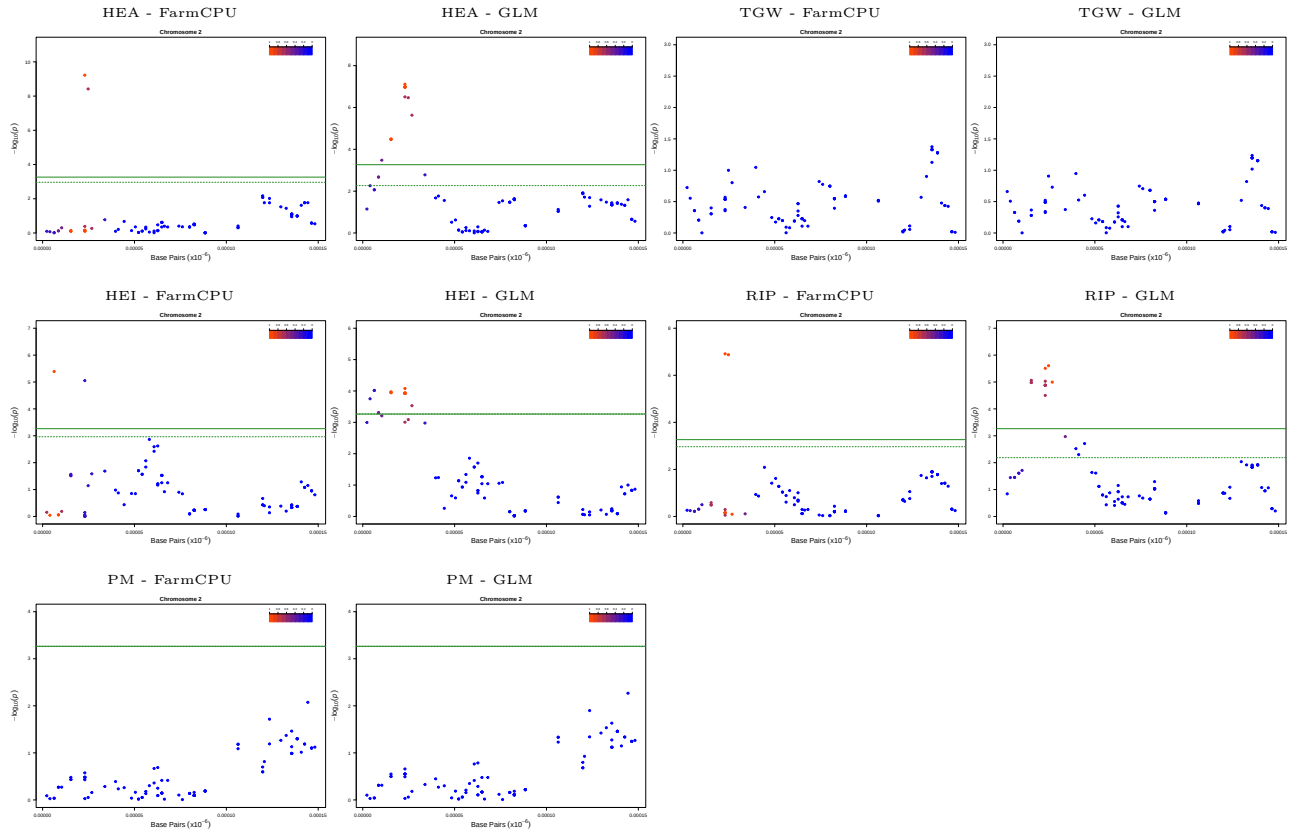


Fig. 5: Manhattan plots of phenotype traits for both models (FarmCPU and GLM).

3.3 Analysis

HEA

- Both model outputs are similar (aligns with QQ)
- FarmCPU identifies 2 Marker to be highly significant with a high pearson r (positional close)
- Same region as GLM

TGW

- Both models outputs identical (aligns with QQ)
- TGW outputs aligns with the expected poly genetic orchestration of this quantitative trait
- Meaning that no marker is surprisingly significant

HEI

- Both model outputs are similar (aligns with QQ)
- FarmCPU identifies 1 Marker to be highly significant with a high pearson r
- FarmCPU identifies 1 Marker to be highly significant but low pearson r
- Same region as GLM

RIP

- Both model outputs are similar (aligns with QQ)
- FarmCPU identifies 2 Marker to be highly significant with a high pearson r (positional close)
- Both Markers seem to align with the markers found in HEA trait
- These markers could correspond to the anti correlation found previously
- Same region as GLM

PM

- Both models outputs identical (aligns with QQ)
- One can clearly see that a wider genome region ($0.10 - 0.15BP(10^{-6})$) has a genetic pattern for the powdery mildew susceptibility (with ok'ish significance $p \approx 0.05$, but not corrected)
- One could dig into this pattern, since it might align with the phenotype distribution (not relevant for this report)

4 Significant Marker

4.1 Linkage Disequilibrium (LD)

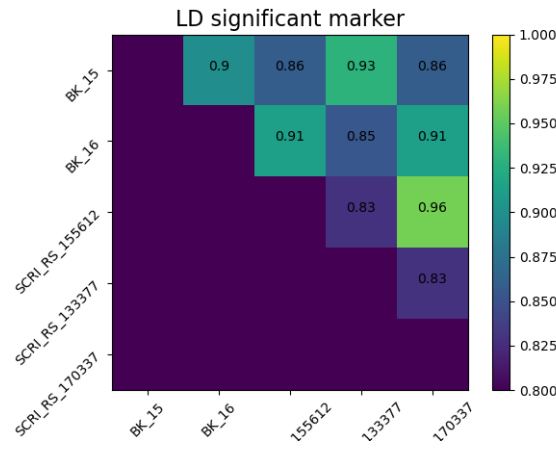


Fig. 6: LD scatter plot of significant markers.

- All significant marker pairs show a $LD > 0.8$
- Quick recap: BK_15, BK_16 and SCRI_RS_170337 (RIP) are extremely close, so high LD is expected

4.2 Phenotype Distribution of Significant Markers w.r.t. quantitative trait

HEA

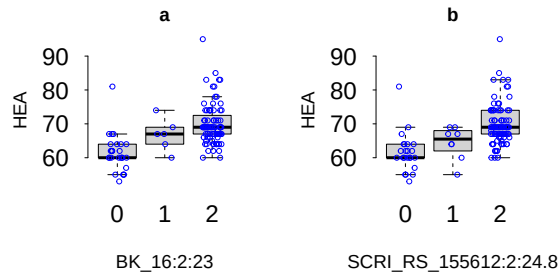


Fig. 7: Phenotype distribution of significant markers w.r.t. quantitative trait. (HEA)

- $\alpha(\text{BK_16}) = +4.2$
- $\alpha(\text{SCRI_RS_155612}) = +1.37$
- Both Marker are separated by 1.8 cM

HEI

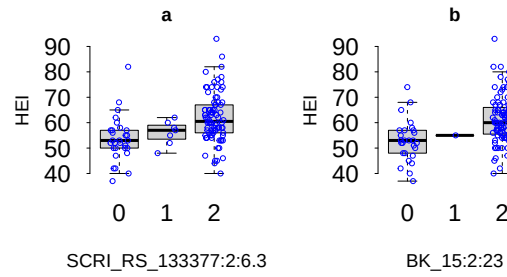


Fig. 8: Phenotype distribution of significant markers w.r.t. quantitative trait. (HEI)

- $\alpha(\text{SCRI_RS_133377}) = +2,62$
- $\alpha(\text{BK_15}) = +4.42$
- Both Markers are separated by 16.7 cM

RIP

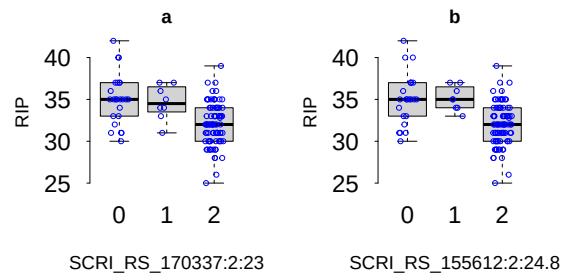


Fig. 9: Phenotype distribution of significant markers w.r.t. quantitative trait. (RIP)

- $\alpha(\text{SCRI_RS_170337}) = -0.59$
- $\alpha(\text{SCRI_RS_155612}) = -1.62$
- Both Marker are separated by 1.8cM

Remark

- BK_16 (HEA) = BK_15 (HEI) = SCRI_RS_170337 (RIP) = 2.3 Mb
→ Selection of these Markers effects 3 traits (pleiotropy)!
- SCRI_RS_155612 (HEA) = SCRI_RS_155612 (RIP) = 2.48 Mb (identical)
→ Selection on this Marker effects 2 traits (pleiotropy)!

5 Marker-assisted selection (trait prediction)

Formula:

$$\hat{\mu} = \mu_A + \alpha_{marker} \cdot SNP$$

- $\hat{\mu}$ = predicted trait value
- μ_A = population mean of genotype A (baseline, SNP = 0)
- α_{marker} = allele effect at the marker (B vs A)
- SNP = genotype coding: 0 (A/A) or 2 (B/B)

Potential Targets

Marker 1	SCRLRS_155612	24.9cM	expected pleiotropic effect
Marker 2	BK_15, BK_16, SCRLRS_170337	23.0cM	expected pleiotropic effect (Multi-Marker complex)
Marker 3	SCRLRS_133377	6.3cM	single trait

Predictions by Marker

	HEA (in days)	HEI (in cm)	RIP (in days)	TGW (in g/1000 grains)	PM (in score)
Marker 1 (SNP=B)	+2.74	–	-3.24	–	–
Marker 2 (SNP=B)	+8.40	+8.84	-1.58	–	–
Marker 3 (SNP=B)	–	+5.24	–	–	–

Note: Multi-marker selection may show additive effects due to high LD between markers.

6 Breeding-Implications

Grain production

- $TGW \times RIP$ correlation = +0.36 (positive, moderate)
- So later ripening $\rightarrow$ higher TGW
- For the RIP markers, α is negative (SCRLRS_155612 = -1.62 , SCRLRS_170337 = -0.59)
- Negative α means B allele $\rightarrow$ earlier ripening (RIP decreases from A to B)
- Earlier ripening $\rightarrow$ lower TGW
- Therefore: A is expected favor TGW (because A keeps RIP later)

Animal Food Production

- One might select for Marker 3 = Allele B, to get higher crops, with an acceptable effect on the other traits (from LD)